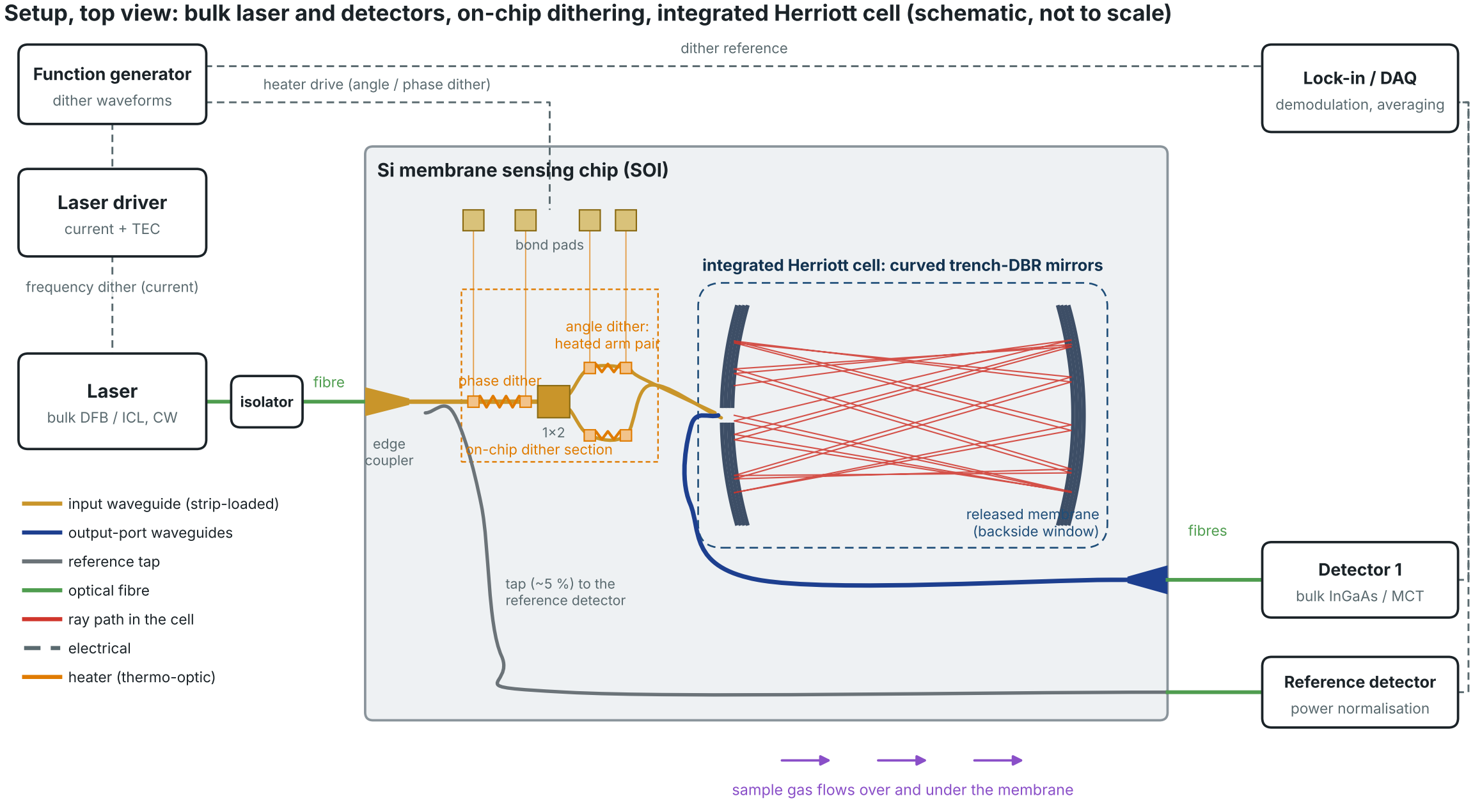
Setup, top view: bulk laser and detectors, on-chip dithering, integrated Herriott cell (schematic, not to scale)
Function generator
dither waveforms
dither reference
heater drive (angle / phase dither)
Lock-in / DAQ
demodulation, averaging
Laser driver
current + TEC
frequency dither (current)
Laser
bulk DFB / ICL, CW
isolator
fibre
Si membrane sensing chip (SOI)
bond pads
integrated Herriott cell: curved trench-DBR mirrors
angle dither:
heated arm pair
phase dither
1×2
on-chip dither section
edge
coupler
released membrane
(backside window)
fibres
tap (~5 %) to the
reference detector
Detector 1
bulk InGaAs / MCT
Reference detector
power normalisation
input waveguide (strip-loaded)
output-port waveguides
reference tap
optical fibre
ray path in the cell
electrical
heater (thermo-optic)
sample gas flows over and under the membrane